



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering

Academic Year 2022 – 2023 (Even Semester)

Name of the Faculty	:	Dr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Course Code and Title	:	ME8691 Computer-Aided Design and Manufacturing
Year and Branch	:	III -Mechanical A
Unit/ Topic	:	III – CAD Standards – Data Exchange Standards
Course Outcome	:	Summarize the different types of Standard systems used in CAD
Topic Learning Outcome	:	Simulate the Data exchange formats of IGES, STEP, STL., using standard CAD models
Activity Chosen		Theory to Practice
Time Allotted for Activity	:	50 minutes
No.of Students participated	:	40
Justification	:	The theoretical content that has been studied during class hours about data formats experienced using the Modern tool of PTC Creo software with standard CAD models, especially for the difference between IGES and STEP.
Details of Implementation	:	Students were assembled in our CAD Laboratory and initially they were exposed to the creation of standard CAD models using Extrude and Revolve commands in the PTC Creo software package. After that, each student converted their CAD part file into IGES & STEP file format. The physical meaning of the difference between these two file formats has been discussed regarding surface and volumetric models.

PO/PSO mapping for the activity

Course Outcome /Programme Outcome	PO1	PO3	PO5	PO9	PSO1
Level of Mapping	2	2	3	2	2
Justification for correlation	To apply the fundamental mechanical engineering concepts to create the cad model	To determine the CAD design objectives, functional requirements and arrive at specifications.	To identify modern engineering tools such as PTC CREO software for engineering activities	To present results as an individual with smooth integration CAD software package	To design component using CAD package

(1 – Low, 2- Moderate, 3-High)

Activity Glimpses





Reflective Critique:

i. Feedback of practice from students and other stakeholders:

Students gave positive oral feedback

ii. Benefit of the practice:

Students learnt to use CAD software package and file format topics.

iii. Whether the practice is adopted in any of the courses early:

Yes, in the same course

Challenges faced in implementation:

Initially students struggled to change the complex geometric file into STEP because of more time consuming process with various factor considerations.

References: Nil

CO3: Students will be able to Summarize the different types of Standard systems used in CAD.

Faculty In Charge

HoD/Mech

